MIL-C-50652/1A (AR)
28 February 1982
SUPERSEDING
MIL-C-50652/1 (MU)
15 April 1971

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548018 (GENERATOR, MODULATOR TRIGGER)

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements, quality assurance provisons, and packaging for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the modulator trigger generator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-50652	Circuit Card Assemblies, Radar Set AN/VPS-2, General Specification for

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

FSC 1285

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

F10548018 - Assembly, Modulator Trigger Generator

PACKAGING DATA SHEET

10548018 - Packaging of Modulator Trigger Generator

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Fabrication. The assembly shall be manufactured in accordance with Drawing F10548018 and drawings pertaining thereto.

3.2 General specifications. The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. As specified in MIL-C-50652.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

3.5 Performance. When supplied with the inputs specified in Table I, the modulator trigger generator shall yield the outputs specified in Table I.

3.6 First Article Inspection. This specification makes provisions for First Article Inspection. Requirements for the submission of First Article Samples by the contractor shall be as specified in the contract.

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3.7 Workmanship. All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface blisters, excess flash, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing the base material. The assembly operations shall be performed in a thorough, workmanlike manner consistent with the best practices of the industry. All required markings shall be neat and sharply defined.

TABLE I. Input and output characteristics

Requirement	Characteristic	Terminals
Input power	5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 300 mA.	P1-30, P1-31 & P1-32 (+5 V) P1-24, P1-25 & P1-26 (+5 V RET)
Input power	25.0 ±0.5 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 100 mA.	P1-55 & P1-56 (+25 V) P1-53 & P1-54 (+25 V RET)
Input signal (PRF)	20 kHz positive pulse of 3.5 ±1.0 volts with a pulse width of NLT 0.5 usec and NMT 0.7 usec wide, measured at the 50% points.	P1-10, P1-51 (SHIELD TIE)
Mod. Trig. Output	20 kHz positive pulse, NLT 10 peak-to-peak, pulse width 0.5 to 0.7 usec at 50% points, and have a rise time less than 100 nsec. The leading edge of pulse shall be delayed from the negative going edge of the input signal by 150 ns maximum with a jumper connected between E18 and E1 on modulator trigger generator.	P1-45, P1-44 (SHIELD TIE)
Mod Trig. Output (delayed)	Same as Mod. Trig. Output except as jumper is moved in steps from E1 toward E17 there will be an additional delay of 8.33 ns ±15% for each step, except for E2 which shall read 10.5 ns ± 15%.	P1-45

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. As specified in MIL-C-50652.

4.3 Inspection provisions. As specified in MIL-C-50652.

4.4 Special sampling. As specified in MIL-C-50652.

4.5 Inspection equipment. As specified in MIL-C-50652.

4.5.1 Government-furnished inspection equipment. As specified in MIL-C-50652.

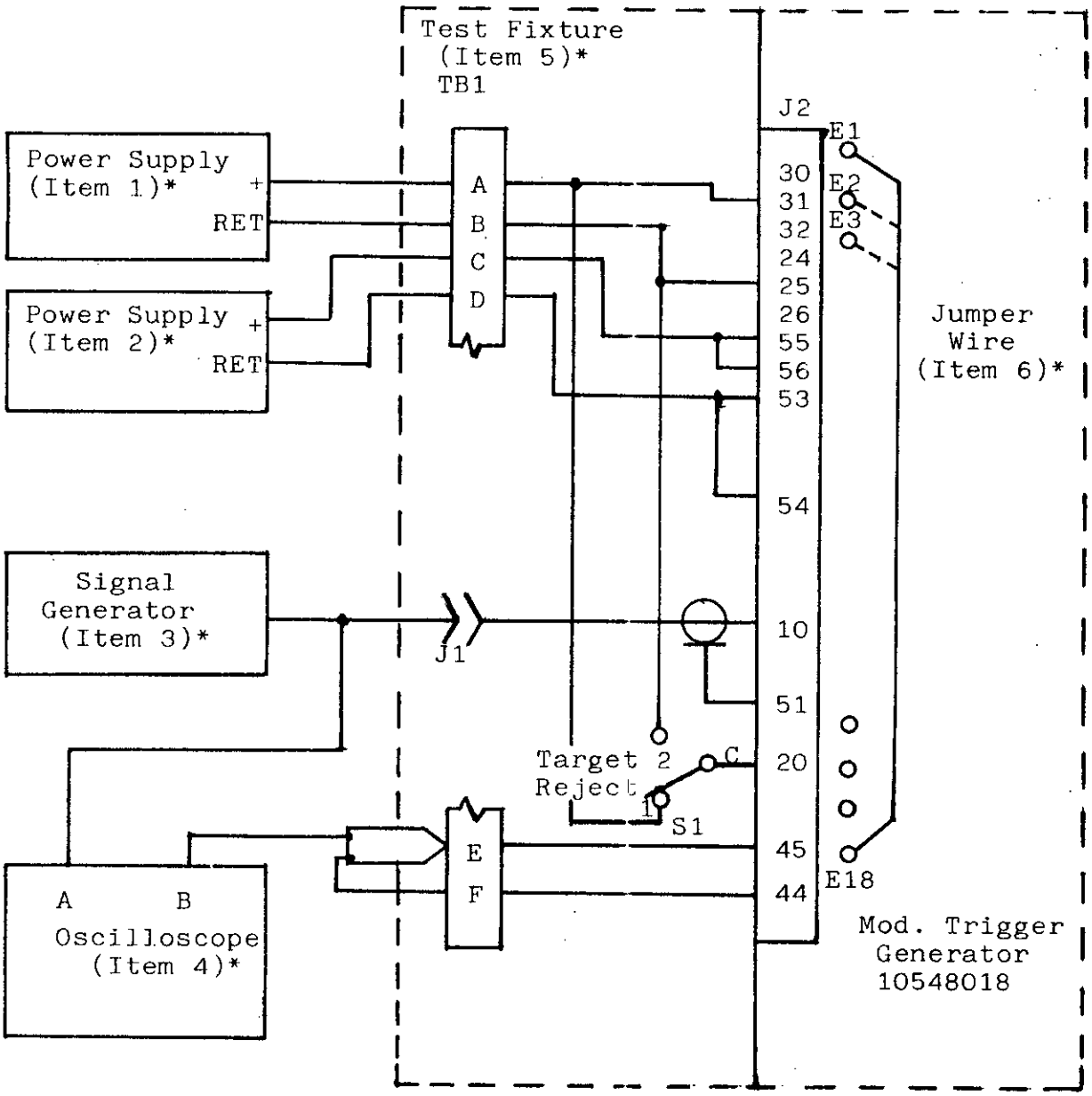
4.5.2 Contractor-furnished inspection equipment. As specified in MIL-C-50652.

4.5.3 Test equipment. In conjunction with 4.5 the following test equipment listed in Table II shall be utilized in the performance of the applicable tests as specified in 4.6.3.

TABLE II. Test equipment list

Item Number		Description
1.	Power supply	5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 300 mA.
2.	Power supply	25.0 ±0.5 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 100 mA.
3.	Signal generator	20 kHz positive pulse of 3.5 ±1.0 volts and the pulse width measurement at the 50% points should be NLT 0.5 usec and NMT 0.7 usec wide.
4.	Oscilloscope	Dual channel; rise time of scope plus measuring probe 10 ns or less. Sweep period: 10 ns per division or greater. Input resistance: greater than 8 megohms. Input capacitance: less than 25 uF. Scope cables: 50 ohms impedance.
5.	Test fixture	As specified in Figure 1. See Table III for test fixture parts list.
6.	Jumper wire	3-inch wire with two minigator clips on each end.

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*Refer to TABLE II.

FIGURE 1. Modulator trigger generator test setup.

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TABLE III. Test fixture parts list

Designation	Nomenclature and description
J1	Connector UG-625B/U
J2	Connector; any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois. (2) 500946-01, National Connector Corp., Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corporation, Willow Grove, Pa.
S1	Switch, SPDT.
TB1	Terminal board, 37TB-6 per MIL-T55164/1 or equivalent.

4.6 Test methods and procedures.

4.6.1 Environmental. As specified in MIL-C-50652.

4.6.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. The modulator trigger generator shall be tested as follows: Adjust power supplies and signal generator, items 1, 2 and 3, to deliver outputs described as "Input power" and "Input signal PRF" in Table I. Connect test equipment, item 1 through 6, as shown in Figure 1. Insert modulator trigger generator into jack J2 in test fixture, item 5.

a. Output signal labeled MOD. TRIG. OUTPUT in Table I shall appear on channel B of oscilloscope, item 4. The measured delay between signal on channel A and channel B shall be as shown in Table I.

b. Move one end of jumper, item 6, in steps from E1 toward E17. The MOD. TRIG. OUTPUT (Delayed) in Table I shall appear on Channel B but there will be an additional delay of 8.33 ns ±15 percent for each step, except for E2 which shall read 10.5 ns ±15%.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548018. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The modulator trigger generator is used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The modulator trigger generator provides a shaped PRF pulse which is applied to the transmitting and receiving systems of the radar set.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable packaging data sheet number (see 5.1).
c. Selection of applicable level of preservation packaging and packing.
d. Level of protection required.
e. Applicable stock number.
f. Requirement for first article submission.

6.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring of the equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAF-I, Dover, NJ 07801.

When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc.).
c. The designs remaining to be submitted and the expected date of transmittal.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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